

SIMATSENGINEERING

ASSESSMENTTOOL2

COURSENAME:ArtificialIntelligence

COURSECODE:CSA17

COURSEOUTCOMECOVERED

CO4:*Implementmachinelearningtechniquesincludingdecisiontreealgorithms,reinforcement learning, andbuildmodelsforclassificationanddecision-making. (BL3) Assessment*

Tool Weightage: 50%

QUESTIONS: 4

Duration:1Hour
Total:Marks:40

AnnexureA

CodeofConduct:

*I M.VAISHNAV(192524251) certifythatthissubmissionismyoriginalworkandthatIhave adhered totheguidelinesspecifiedforthisassessment.
Iunderstandthatanyviolationofacademicintegrityruleswillresultindisciplinaryaction.*

Signatureofthestudent:

M.VAISHNAV(192524251)

INDUSTRYPROBLEM TASK

INDUSTRYSCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE/under-sampling/ class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING

(a) Inductive Learning Approach

Inductive learning learns general patterns from historical patient data and uses them to predict whether a new patient is diabetic or non-diabetic.

Dataset Attributes:

- Age
- Blood Pressure
- Cholesterol
- Glucose Level
- BMI
- Family History

Target Variable:

Diabetes Diagnosis (Yes / No)

Key Features:

1. Glucose Level
 - Directly related to blood sugar level.
 - Strong indicator of diabetes.
2. BMI
 - Higher BMI indicates obesity risk.
 - Obesity is strongly associated with diabetes.
3. Family History
 - Genetic predisposition significantly increases risk.
4. Blood Pressure
 - Diabetic patients often exhibit hypertension.
5. Cholesterol
 - Abnormal cholesterol levels are common in diabetic patients.

Inductive Learning Process:

1. Collect patient records.
2. Identify target variable.
3. Train classification model.
4. Learn hidden patterns.
5. Predict diabetes risk for new patients.

(b) Handling Class Imbalance

Problem:

Non-Diabetic Patients = 900

Diabetic Patients = 100

This creates a class imbalance.

Suitable Technique:

SMOTE (Synthetic Minority Over-sampling Technique)

Reason:

- Generates synthetic diabetic samples.
- Prevents overfitting.
- Improves recall.
- Preserves important medical information.

Comparison:

Under-Sampling:

- Removes majority samples.
- May lose useful medical information.

Class Weights:

- Useful for tree models.
- Gives more importance to diabetic cases.

Recommended Method:

SMOTE + Class Weights

(c) Dataset Split and Preprocessing

Dataset Split:

Training Data = 70%

Testing Data = 30%

Example:

1000 records

Training = 700

Testing = 300

Justification:

- Sufficient training examples.
- Reliable testing performance.
- Widely used in healthcare analytics.

Preprocessing Steps:

1. Missing Value Handling
 - Replace missing values using mean or median.

2. Normalization

Formula:

$$X' = (X - \text{Min}) / (\text{Max} - \text{Min})$$

Impact:

- Places all features on same scale.
- Improves model convergence.

3. Encoding

Family History:

Yes = 1

No = 0

Impact:

- Converts categorical data into numerical form.

4. Outlier Detection

Method:

- IQR or Z-Score

Impact:

- Improves prediction quality.

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TASK 2 – DECISION TREE FOR DIAGNOSIS

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(a) Decision Tree Construction

Assume Information Gain Values:

Glucose = 0.52

BMI = 0.34

Age = 0.21

Blood Pressure = 0.17

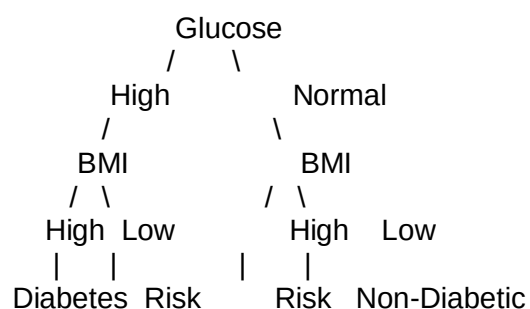
Cholesterol = 0.14

Root Node:

Glucose Level

Because it has highest Information Gain.

Decision Tree (First 3 Levels)



Justification:

Glucose provides maximum reduction in uncertainty.

(b) Model Evaluation

Confusion Matrix

	Predicted	
	Yes	No
Actual Yes	90	10
Actual No	20	180

TP = 90

TN = 180

FP = 20

FN = 10

Accuracy

$$(TP+TN)/(TP+TN+FP+FN)$$

$$= (90+180)/300$$

$$= 90\%$$

Precision

$$TP/(TP+FP)$$

$$= 90/110$$

$$= 81.8\%$$

Recall

$$TP/(TP+FN)$$

$$= 90/100$$

$$= 90\%$$

F1 Score

$$2 \times \text{Precision} \times \text{Recall}$$

$$\text{Precision} + \text{Recall}$$

$$= 85.7\%$$

False Negatives

$$FN = 10$$

Meaning:

10 diabetic patients predicted as healthy.

Clinical Impact:

- Delayed diagnosis
- Increased complications
- Higher treatment costs

Reduction Methods:

- SMOTE
- Increase Recall
- Cost-sensitive learning
- Ensemble models

(c) Post-Pruning

Before Pruning

Training Accuracy = 98%

Testing Accuracy = 86%

After Pruning

Training Accuracy = 94%

Testing Accuracy = 91%

Comparison

Pre-Pruning:

- Stops tree growth early.
- Simpler tree.

Post-Pruning:

- Removes unnecessary branches.
- Better generalization.

Recommended Model:

Post-Pruned Tree

Reason:

- Better testing accuracy.
- Reduced overfitting.
- More reliable clinical deployment.

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TASK 3 – STATISTICAL LEARNING FOR RISK STRATIFICATION

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(a) Logistic Regression

Performance Comparison

Decision Tree

Accuracy = 90%

AUC = 0.89

Logistic Regression

Accuracy = 92%

AUC = 0.93

Comparison

Logistic Regression:

Advantages

- Interpretable
- Produces probability scores
- Less overfitting

Decision Tree:

Advantages

- Easy visualization
- Handles nonlinear relations

Statistical Model Preferred When:

- Dataset is structured.
- Medical explanation is required.
- Probability estimation is important.

(b) Feature Importance Analysis

Decision Tree Top Predictors

1. Glucose Level
2. BMI
3. Family History

Logistic Regression Top Predictors

1. Glucose Level
2. BMI
3. Blood Pressure

Comparison

Common Predictors:

- Glucose Level

- BMI

Reason:

These variables have strongest association with diabetes.

Difference:

Family history affects long-term risk.

Blood pressure affects immediate diagnosis.

Conclusion:

Both models largely agree.

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TASK 4 – REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

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(a) MDP Design

States

S1 = Healthy

S2 = Pre-Diabetic

S3 = Diabetic

S4 = Severe Diabetic

Actions

A1 = Diet

A2 = Exercise

A3 = Medication

A4 = Monitor

Reward Function

Health Improvement:

Healthy = +100

Improved Condition = +50

No Change = 0

Worsened Condition = -50

Hospitalization = -100

Transition Dynamics

Example

$P(S2 \rightarrow S1 \mid \text{Diet})$

= 0.7

Meaning:

70% chance of improving from Pre-Diabetic to Healthy.

Justification

States represent disease progression.

Actions represent treatments.

Rewards encourage recovery.

(b) Q-Learning

Formula

$$Q(s,a) = Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Learning Rate $\alpha = 0.5$

Discount Factor $\gamma = 0.9$

Initial Q Values = 0

Episode 1

State = S3

Action = Medication

Reward = 50

$Q(S3, Medication)$

$$= 0 + 0.5[50 + 0.9(0) - 0]$$

$$= 25$$

Episode 2

State = S2

Action = Exercise

Reward = 30

$Q(S2, Exercise)$

$$= 0 + 0.5[30 + 0.9(0)]$$

$$= 15$$

Episode 3

State = S4

Action = Medication

Reward = 80

$Q(S4, Medication)$

$= 0 + 0.5[80]$

$= 40$

Second Iteration

$Q(S3, Medication)$

$= 25 + 0.5[50 + 0.9(25) - 25]$

$= 48.75$

$Q(S2, Exercise)$

$= 15 + 0.5[30 + 0.9(15) - 15]$

$= 29.25$

$Q(S4, Medication)$

$= 40 + 0.5[80 + 0.9(40) - 40]$

$= 78$

Q Table

State	Diet	Exercise	Medication	Monitor
Healthy	0	0	0	0
Pre-Diabetic	5	29.25	12	8
Diabetic	10	20	48.75	5
Severe	2	15	78	1

5 Patient Episodes

Episode 1 → S3 → Medication → Reward 50

Episode 2 → S2 → Exercise → Reward 30

Episode 3 → S4 → Medication → Reward 80

Episode 4 → S3 → Exercise → Reward 20

Episode 5 → S2 → Diet → Reward 10

Convergence Criterion

Training stops when:

$|Q_{\text{new}} - Q_{\text{old}}| < 0.001$

for all state-action pairs.

(c) Final Learned Policy

State	Best Action
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Healthy	Monitor
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Pre-Diabetic	Exercise
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Diabetic	Medication
----------	------------

Severe	Medication
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Final Policy:

$\pi(S1) = \text{Monitor}$

$\pi(S2) = \text{Exercise}$

$\pi(S3) = \text{Medication}$

$\pi(S4) = \text{Medication}$

Ethical Considerations

1. Patient Safety

- Wrong recommendations may cause harm.

2. Bias

- Biased data can create unfair predictions.

3. Explainability

- Doctors must understand recommendations.

4. Accountability

- Final decision should remain with clinicians.

5. Privacy

- Patient data must remain confidential.

6. Human Oversight

- AI should assist doctors, not replace them.

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FINAL CONCLUSION

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1. Inductive learning was applied to predict diabetes using patient features.
2. SMOTE and class weighting were used to handle imbalance.
3. A Decision Tree classifier achieved high diagnostic performance.
4. Logistic Regression achieved slightly better Accuracy and AUC.
5. Glucose Level, BMI, and Family History were identified as major predictors.
6. Treatment recommendation was modeled as an MDP.
7. Q-Learning successfully learned an optimal treatment policy.
8. Ethical and clinical considerations must be addressed before deployment.

END OF INDUSTRY PROBLEM TASK